

Question One: (15 Marks) *Start a new sheet of paper.*

a) Find $\int \frac{x}{\sqrt{2-x^2}} dx$ using the substitution $x = \sqrt{2} \sin \theta$. [2]

b) Show that $\sin(A+B) + \sin(A-B) = 2 \sin A \cos B$, and hence find

$$\int \sin 5x \cos 3x dx. \quad [3]$$

c) Use Integration by Parts to show that $\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$. [3]

d) Given that $J_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$:

i) Prove that $J_n = \frac{(n-1)}{n} J_{n-2}$, where n is an integer and $n \geq 2$. [4]

ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x dx$. [3]

Question Two: (15 Marks) *Start a new sheet of paper.*

a) Given that $z = 2 + i$ and $\omega = 2 - 3i$, find, in the form $a + ib$

i) $\left(\frac{\bar{z}}{z}\right)^2$ [1]

ii) $\left(\frac{z}{\omega}\right)$ [1]

b) On the Argand diagram, shade the region where the inequalities

$$-1 < |z| < 1 \text{ and } \frac{\pi}{4} < \arg(z) < \frac{3\pi}{4} \text{ both hold.} \quad [3]$$

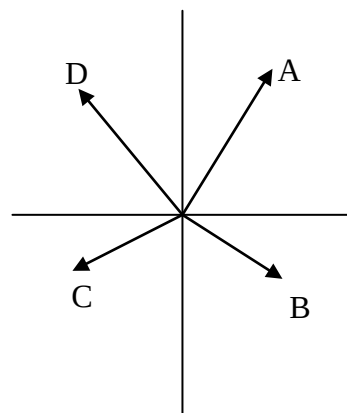
c) Find the complex square roots of $7 + 6i\sqrt{2}$, giving your answer in the form $a + ib$, where a and b are real. [3]

(Question 2 continued over)

d) Given the two complex numbers $z_1 = r_1 \text{cis} \theta$ and $z_2 = r_2 \text{cis} \phi$,

i) Show that, if z_1 and z_2 are parallel, $z_1 = kz_2$, for k real. [1]

ABCD is a quadrilateral with vertices A, B, C and D represented by the complex numbers (vectors) z_1, z_2, z_3 and z_4 , as shown in the sketch opposite.



ii) Give two possible vectors (in terms of z_1, z_2) for side AB. [1]

iii) If ABCD is a parallelogram, show that $z_1 - z_2 - z_3 + z_4 = 0$. [3]

e) Explain the fallacy in the following argument: [2]

$$-1 = \sqrt{-1} \times \sqrt{-1} = \sqrt{(-1)(-1)} = \sqrt{1} = 1. \text{ Hence } 1 = -1.$$

Question Three: (15 Marks) *Start a new sheet of paper.*

a) $F(x)$ is defined by the equation $f(x) = x^2 \left(x - \frac{3}{2} \right)$, on the domain $-2 \leq x \leq 2$.

Note: each sketch below should take about one third of a page.

i) Draw a neat sketch of $F(x)$, labelling all intersections with coordinate axes and turning points. [2]

ii) Sketch $y = \frac{1}{F(x)}$ [2]

iii) Sketch $y = \sqrt{F(x)}$ [2]

iv) Sketch $y = \ln(F(|x|))$ [2]

b) The Hyperbola $\mathcal{H}$ has the equation $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

i) Find the eccentricity of $\mathcal{H}$. [1]

ii) Find the coordinates of the foci of $\mathcal{H}$. [1]

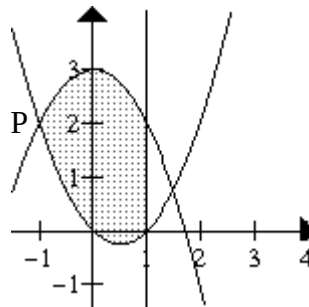
iii) Draw a neat one third of a page size sketch of $\mathcal{H}$. [2]

iv) The line $x = 6$ cuts $\mathcal{H}$ at A and B. Find the coordinates of A and B if A is in the first quadrant. [1]

v) Derive the equation of the tangent to $\mathcal{H}$ at A. [2]

Question Four: (15 Marks) *Start a new sheet of paper.*

a) The shaded region bounded by $y = 3 - x^2$, $y = x^2 - x$ and $x = 1$ is rotated about the line $x = 1$. The point P is the intersection of $y = 3 - x^2$ and $y = x^2 - x$ in the second quadrant.



i) Find the x coordinate of P. [1]

ii) Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral. [3]

iii) Evaluate the integral in part (ii) above. [2]

(Question 4 continued over)

b) Find real numbers A, B and C such that

$$\frac{x}{(x-1)^2(x-2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x-2)}. \quad [2]$$

Hence show that $\int_0^{\frac{1}{2}} \frac{x}{(x-1)^2(x-2)} dx = 2 \ln\left(\frac{3}{2}\right) - 1.$ [2]

c) Find all x such that $\cos 2x = \sin 3x$, if $0 \leq x \leq \frac{\pi}{2}$. [2]

d) Solve for x : $\tan^{-1}(3x) - \tan^{-1}(2x) = \tan^{-1}\left(\frac{1}{5}\right)$ [3]

Question Five: (15 Marks) *Start a new sheet of paper.*

a) For the polynomial equation $x^3 + 4x^2 + 2x - 3 = 0$ with roots α, β and γ , find:

i) The value of $\alpha^2 + \beta^2 + \gamma^2$ [1]

ii) The equation whose roots are $(1-\alpha), (1-\beta), (1-\gamma)$. [2]

iii) The equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. [3]

b) Determine all the roots of $8x^4 - 25x^3 + 27x^2 - 11x + 1 = 0$ given that it has a root of multiplicity 3. [4]

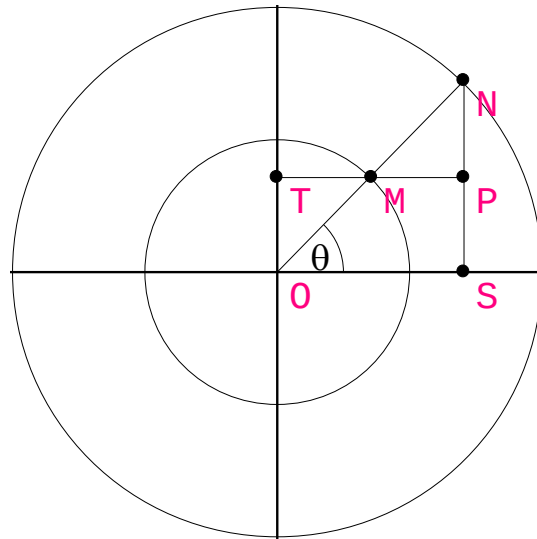
c) The equation $x^4 + 4x^3 + 5x^2 + 2x - 20 = 0$ has roots α, β, γ and δ over the complex field.

i) Show that the equation whose roots are $\alpha + 1, \beta + 1, \gamma + 1$ and $\delta + 1$ is given by $x^4 - x^2 - 20 = 0$. [2]

ii) Hence solve the equation $x^4 + 4x^3 + 5x^2 + 2x - 20 = 0$. [3]

Question Six: (15 Marks) *Start a new sheet of paper.*

a)



The circles above have centres at O and radii of 5 units and 3 units respectively.

A ray from O making an angle θ with the positive x -axis, cuts the circles at the points M and N as shown.

NS is drawn parallel to the y -axis and MT parallel to the x -axis.

NS and MT intersect at P.

- i) Show that the parametric equations of the locus of P in terms of θ are given by $x = 5 \cos \theta$ and $y = 3 \sin \theta$. [2]
- ii) By eliminating θ , find the Cartesian equation of this locus. [1]
- iii) Find the equation of the normal (in general form) at the point P when $\theta = \frac{\pi}{3}$. [2]

b) The functions $S(x)$ and $C(x)$ are defined by the formulae

$$s(x) = \frac{1}{2}(e^x - e^{-x}) \text{ and } c(x) = \frac{1}{2}(e^x + e^{-x}).$$

- i) Verify that $S'(x) = C(x)$. [1]

ii) Show that $S(x)$ is an increasing function for all real x . [1]

iii) Prove $[C(x)]^2 = 1 + [S(x)]^2$ [2]

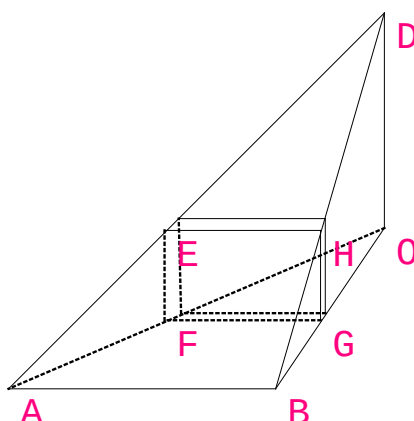
iv) $S(x)$ has an inverse function, $S^{-1}(x)$, for all real values of x .
Briefly justify this statement. [1]

v) Let $y = S^{-1}(x)$. Prove that $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}$. [2]

vi) Hence, or otherwise, show that $S^{-1}(x) = \ln\{x + \sqrt{1+x^2}\}$. [3]

Question Seven: (15 Marks) *Start a new sheet of paper.*

a) Let OAB be an isosceles triangle, $OA = OB = r$, $AB = b$.



Let OABD be a triangular pyramid with height $OD = h$ and OD perpendicular to the plane of OAB as in the diagram above.

Consider a slice S of the pyramid of width δa as shown at $EFGH$ in the diagram. The slice S is perpendicular to the plane of OAB at FG with $FG \parallel AB$ and $BG = a$. Note that $GH \parallel OD$.

i) Show that the volume of S is $\left(\frac{r-a}{r}\right)b\left(\frac{ah}{r}\right)\delta a$ when δa is small.
(You may assume the slice is approximately a rectangular prism of base $EFGH$ and height δa). [3]

ii) Hence show that the pyramid $DOAB$ has a volume of $\frac{1}{6}hbr$. [2]

iii) Suppose now that $\angle AOB = \frac{2\pi}{n}$ and that n identical pyramids DOAB are arranged about O as the centre with common vertical axis OD to form a solid C. Show that the volume V_n of C is given by $V_n = \frac{1}{3}r^2hn\sin\frac{\pi}{n}$. [2]

iv) Note that when n is large, C approximates a right circular cone. Hence, find $\lim_{n \rightarrow \infty} V_n$ and verify a right circular cone of radius r and height h has a volume of $\frac{1}{3}\pi r^2h$. [2]

b) On the hyperbola $xy = c^2$, three points P, Q and R are on the same branch, with parameters p, q and r respectively. The tangents at P and Q intersect at U. If O, U and R are collinear, find the relationship between p, q and r . [6]

Question Eight: (15 Marks) *Start a new sheet of paper.*

a)

i) Use the substitution $x = \frac{2}{3}\sin\theta$ to prove that $\int_0^{\frac{2}{3}} \sqrt{4-9x^2} dx = \frac{\pi}{3}$. [3]

ii) Hence, or otherwise, find the area enclosed by the ellipse

$$9x^2 + y^2 = 4. \quad [1]$$

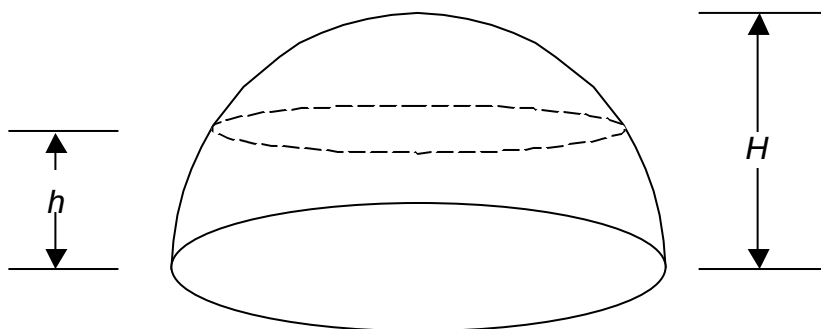
b)

i) Use an appropriate substitution to verify that $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$. [2]

ii) Deduce that the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is given

by πab . [2]

- c) The diagram below shows a mound of height H . At height h above the horizontal base, the horizontal cross-section of the mound is elliptical in shape, with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2$, where $\lambda = 1 - \frac{h^2}{H^2}$, and x, y are appropriate coordinates in the plane of the cross-section.



Show that the volume of the mound is $\frac{8\pi abH}{15}$. [3]

- d) The quadratic equation $x^2 - (2\cos\theta)x + 1 = 0$ has roots α and β .

i) Find expressions for α and β . [1]

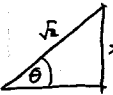
ii) Show that $\alpha^{10} + \beta^{10} = 2\cos(10\theta)$. [3]

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QUESTION ONE:

a) $x = \sqrt{2} \sin \theta$ so $dx = \sqrt{2} \cos \theta d\theta$

$$\begin{aligned} & \int \frac{x}{\sqrt{2-x^2}} dx \\ &= \int \frac{\sqrt{2} \sin \theta \cdot \sqrt{2} \cos \theta d\theta}{\sqrt{2-(2 \sin^2 \theta)}} \\ &= \int \frac{2 \sin \theta \cos \theta d\theta}{\sqrt{2(1-\sin^2 \theta)}} \\ &= \int \frac{\sqrt{2} \sin \theta \cos \theta d\theta}{\cos \theta} \\ &= \int \sqrt{2} \sin \theta d\theta \\ &= -\sqrt{2} \cos \theta + c \\ &= -\sqrt{2} \cdot \frac{\sqrt{2-x^2}}{\sqrt{2}} + c \\ &= -\sqrt{2-x^2} + c \end{aligned}$$

also 
with $\sin \theta = \frac{x}{\sqrt{2}}$
 $x^2 + y^2 = 2$
 $y^2 = 2 - x^2$
 $y = \sqrt{2-x^2}$
 $\therefore \cos \theta = \frac{\sqrt{2-x^2}}{\sqrt{2}}$

b) $\sin(A+B) = \sin A \cos B + \sin B \cos A$ -①
 $\sin(A-B) = \sin A \cos B - \sin B \cos A$ -②
 $\sin(A+B) + \sin(A-B) = 2 \sin A \cos B$ ①+②

$$\begin{aligned} & \int \sin 5x \cos 3x dx \\ &= \int \frac{1}{2} (\sin(5x+3x) + \sin(5x-3x)) dx \\ &= \int \frac{1}{2} \sin 8x + \frac{1}{2} \sin 2x dx \\ &= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + c \end{aligned}$$

c) $\int_0^1 \tan^{-1} x dx$
 $= \int_0^1 \frac{d(x)}{dx} \cdot \tan^{-1} x dx$
 $= [x \tan^{-1} x]_0^1 - \int_0^1 x \cdot \frac{1}{1+x^2} dx$
 $= (\frac{\pi}{4} - 0) - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$
 $= \frac{\pi}{4} - \frac{1}{2} [\ln(1+x^2)]_0^1$
 $= \frac{\pi}{4} - \frac{1}{2} (\ln 2 - \ln 1)$
 $= \frac{\pi}{4} - \frac{1}{2} \ln 2$

MARKING

COMMENTS

P1

① converting from x to θ , & resolving for $\cos \theta$

① answer

① showing relationship

① correct use of formula

① Answer

① correct integration by parts

① correct log integration

① correct algebra

• some got $dx = \sqrt{2} \cos \theta d\theta$, but the left it out of the substitution when changing to θ !!
• very common error was not expressing the indefinite integral back in terms of x ;
 $-\sqrt{2} \cos \theta + c$ got 1 mark.
 $-\sqrt{2} \cos(\sin^{-1}(\frac{x}{\sqrt{2}})) + c$ was also not sufficient for both marks (not simplest form).

• well done

• well done

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d) i) $J_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$
 $= \int_0^{\frac{\pi}{2}} \cos x \cos^{n-1} x dx$
 $= [\sin x \cos^{n-1} x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1) \cos^{n-2} x \cdot \sin x \cdot \sin x dx$
 $= (0-0) + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \sin^2 x dx$
 $= (n-1) \int_0^{\frac{\pi}{2}} (1-\cos^2 x) \cos^{n-2} x dx$
 $= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \cos^n x dx$

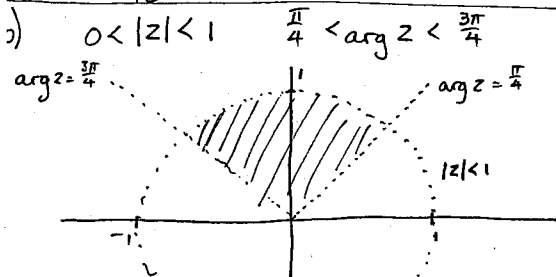
$$\begin{aligned} \therefore J_n &= (n-1) J_{n-2} - (n-1) J_n \\ \therefore J_n + (n-1) J_n &= (n-1) J_{n-2} \\ n J_n &= (n-1) J_{n-2} \\ \therefore J_n &= \frac{(n-1)}{n} J_{n-2} \end{aligned}$$

ii) $\int_0^{\frac{\pi}{2}} \cos^6 x dx = J_6$
 $\therefore J_6 = \frac{5}{6} \cdot \frac{3}{4} \cdot J_4$
 $= \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} J_2$
 $= \frac{15}{48} \cdot \int_0^{\frac{\pi}{2}} dx$
 $= \frac{15}{48} \cdot [x]_0^{\frac{\pi}{2}}$
 $= \frac{15\pi}{96}$

QUESTION TWO:

i) a) $(\bar{z})$
 $= (2-i)^2$
 $= 4 - 4i - 1$
 $= 3 - 4i$

ii) $(\frac{z}{w})$
 $= \frac{(2+i) \cdot (2+3i)}{(2-3i) \cdot (2+3i)}$
 $= \frac{4+6i+2i-3}{4+6i+2i-3}$
 $= \frac{1+8i}{1+8i}$



MARKING

COMMENTS

P1

① correct method for splitting as

① resolving to integral

① resolving to J_{n-2}

① correct algebra to solution

① correctness of formula

① evaluates J_0

① answer

① answer

① answer

① boundaries

① correct $|z|$

① correct arg limits

• many tried the approach $\int \frac{d(x)}{dx} \cdot \cos^n x dx$ and got lost. These need to be known.

• well done.

• some simple mistakes made

• generally good

• generally good.

$$c) (a+ib) = \sqrt{7+6i\sqrt{2}} \quad a, b \text{ real}$$

$$\therefore (a+ib)^2 = 7+6i\sqrt{2}$$

$$a^2 - b^2 + 2abi = 7 + 6i\sqrt{2}$$

equating real and imaginary parts.

$$a^2 - b^2 = 7 \quad \text{--- (1)} \quad 2ab = 6\sqrt{2}$$

$$\therefore a = \frac{6\sqrt{2}}{2b} = \frac{3\sqrt{2}}{b} \quad \text{--- (2)}$$

substituting (2) in (1):

$$\left(\frac{3\sqrt{2}}{b}\right)^2 - b^2 = 7$$

$$\frac{18}{b^2} - b^2 = 7$$

$$18 - b^4 = 7b^2$$

$$\text{ie } b^4 + 7b^2 - 18 = 0$$

$$\therefore (b^2 + 9)(b^2 - 2) = 0$$

$$b^2 = 2, -9$$

reject $b^2 = -9$ as b is real.

$$\therefore b = \pm\sqrt{2} \quad \text{in (2):}$$

$$b = \sqrt{2}$$

$$b = -\sqrt{2}$$

$$a = \frac{3\sqrt{2}}{\sqrt{2}}$$

$$a = \frac{3\sqrt{2}}{-\sqrt{2}}$$

$$= 3$$

$$= -3$$

$$\therefore \text{roots are } 3 + \sqrt{2}i, -3 - \sqrt{2}i$$

$$d) \therefore \text{for } z_1 \parallel z_2, \theta = \phi, \therefore z_2 = r_2 \cos \theta$$

$$\therefore \cos \theta = \frac{z_1}{r_2}$$

$$\therefore \text{from } z_1 = r_1 \cos \theta$$

$$= r_1 \cdot \frac{z_2}{r_2}$$

$$\therefore z_1 = k z_2 \quad \text{where } k = \frac{r_1}{r_2}$$

ii) Side AB is either

$\vec{AB}$

or

$\vec{BA}$

$$OA + AB = OB$$

$$OB + BA = OA$$

$$\therefore AB = OB - OA$$

$$\therefore BA = OA - OB$$

$$= z_2 - z_1$$

$$= z_1 - z_2$$

$$= -(z_2 - z_1)$$

① setup a, b relationship

• mostly well done. Some students tried to use formulas for finding square roots of complex numbers (not very successfully)

① resolve for correct b values

① correct roots.

① deducing relationship.

① for both possibilities

• OK

iii) side CD is given by $z_3 - z_4$ or $-(z_4 - z_3)$

as $CD \parallel AB$, $(z_4 - z_3) = k(z_2 - z_1)$

but opposite sides of a parallelogram are equal,

$$\text{so } |z_4 - z_3| = |z_2 - z_1| \Rightarrow k = \pm 1$$

$$k=1: \therefore z_4 - z_3 = z_2 - z_1$$

$$\text{or } z_1 - z_2 - z_3 + z_4 = 0$$

$$k=-1: \therefore z_4 - z_3 = -(z_2 - z_1)$$

but from (ii), AB (or BA) can be

either $z_2 - z_1$ or $-(z_2 - z_1)$

$$\therefore z_4 - z_3 = z_2 - z_1$$

$$\therefore z_1 - z_2 - z_3 + z_4 = 0$$

iv) the argument uses only real numbers, not complex numbers, thus

$$-1 = (a+ib)^2, \text{ where } a, b \text{ are real.}$$

$$= a^2 - b^2 + 2abi$$

$$\therefore a^2 - b^2 = -1 \text{ and } 2ab = 0$$

$$b=0 \Rightarrow a^2 = -1, \text{ which cannot happen as } a \text{ is real}$$

$$\text{so } a=0 \text{ and } -b^2 = -1 \Rightarrow b = \pm 1$$

$\therefore$ the roots are $-i$ and i

$$i^2 = -1 \text{ and } (-i)^2 = (-1)^2 i^2 = -1$$

$$\text{fn. } \sqrt{-1} \times \sqrt{-1} = \sqrt{-1} \times \sqrt{-1} = i \times i \neq \sqrt{(-1) \times (-1)},$$

① side CD

① deriving k

① both cases for k .

• not well done.

① identifies use of real nos to try and solve a complex no problem

① demonstrates correct procedure (in some way).

• Some good answers, most realized something was wrong, but couldn't articulate what it was

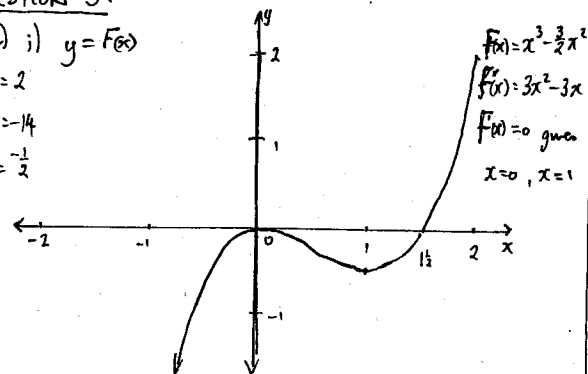
QUESTION 3:

$$a) i) y = f(x)$$

$$f(x) = 2$$

$$f(-2) = -14$$

$$f(1) = -\frac{1}{2}$$

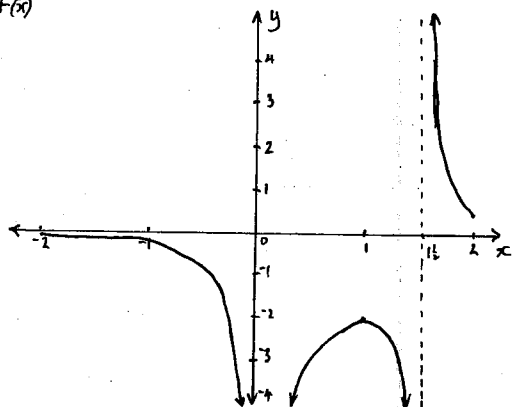


① axis intercepts

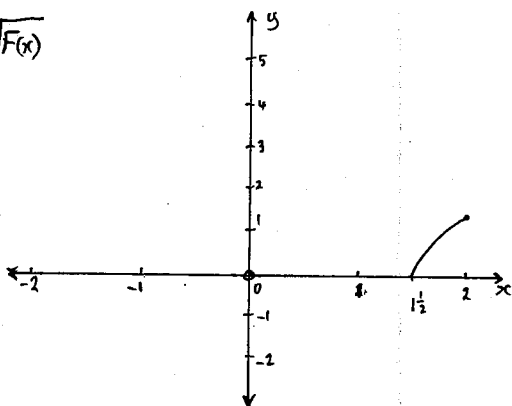
① turning pt. @ $(1, -\frac{1}{2})$

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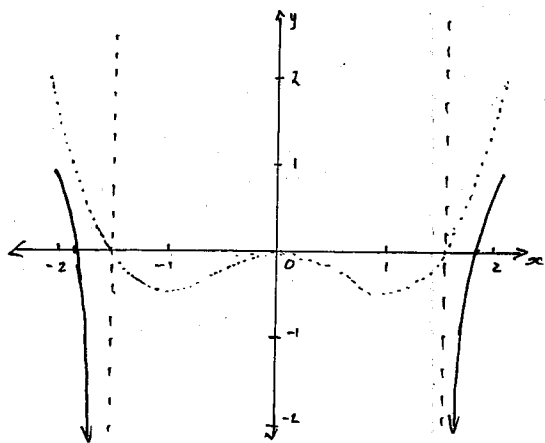
ii) $y = \frac{1}{F(x)}$



iii) $y = \sqrt{F(x)}$



iv)



MARKING

COMMENTS

① asymptotes

① correct shape in areas.

① correct intercept

① correct shape in areas

no penalty for (0,0) not plotted

① asymptotes

① correct shape in area.

p5 SOLUTIONS: Y12 TRIAL HSC. EXTN II: 2003

b) i) $a=5$, $b=3$ and for hyperbola: $b^2 = a^2(e^2 - 1)$

$\therefore 9 = 25(e^2 - 1)$

$e^2 - 1 = \frac{9}{25}$

$e^2 = \frac{34}{25}$

$\therefore e = \frac{\sqrt{34}}{5}$

ii) Foci are $S(ae, 0)$ and $S'(-ae, 0)$

$\therefore (\sqrt{34}, 0)$ and $(-\sqrt{34}, 0)$

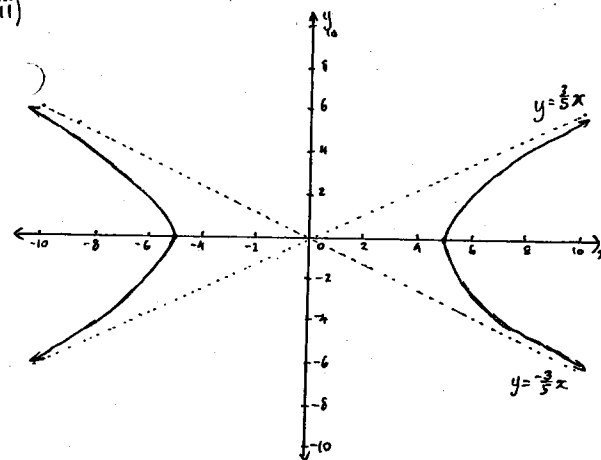
MARKING

COMMENTS

① correct value of e

① correct foci.

iii)



① asymptotes

① correct shape + intercepts.

could also get this mark if shape is reasonable & scale indicated on y-axis

iv) $x=6$: $\frac{36}{25} - \frac{y^2}{9} = 1$

$\frac{36}{25} - 1 = \frac{y^2}{9}$

$y^2 = \frac{9 \times 11}{25}$

$\therefore y = \pm \frac{\sqrt{99}}{5}$

$\therefore A$ is $(6, \frac{\sqrt{99}}{5})$ and B is $(6, -\frac{\sqrt{99}}{5})$

v) $\frac{x^2}{25} - \frac{y^2}{9} = 1$

$\therefore \frac{2x}{25} - \frac{2y}{9} \frac{dy}{dx} = 0$

so $\frac{dy}{dx} = \frac{-2x}{25} \cdot \frac{9}{2y}$

$= \frac{9x}{25y}$

at $(6, \frac{\sqrt{99}}{5})$, $\frac{dy}{dx} = \frac{9 \cdot 6 \cdot 5}{25 \cdot \sqrt{99}}$

$= \frac{54}{5\sqrt{99}}$

① correct A and B

① correct differentiation for $\frac{dy}{dx}$

① correct subst to eqn.

ignored small arithmetic errors.

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v) (cont) $\therefore y - \frac{\sqrt{99}}{5} = \frac{54}{5\sqrt{99}}(x-6)$
 $5\sqrt{99}y - 99 = 54x - 324$
 $0 = 54x - 5\sqrt{99}y - 225$

QUESTION 4:

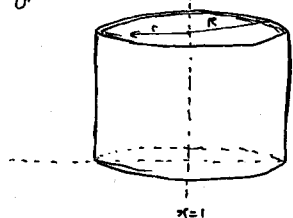
a) i) at P: $3-x^2 = x^2 - x$

$\therefore 0 = 2x^2 - x - 3$
 $= 2x^2 + 2x - 3x - 3$
 $= 2x(x+1) - 3(x+1)$
 $= (x+1)(2x-3)$

$\therefore x = -1, \frac{3}{2}$

$\therefore x$ co-ord of P is -1 (as P is in 2nd quadrant)

ii) typical shell:



inner radius: $r = 1-x$
 outer Radius: $R = 1-(x+\delta x)$
 $\therefore$ Area of annulus:
 $\delta A = \pi R^2 - \pi r^2$
 $= \pi(1-(x+\delta x))^2 - \pi(1-x)^2$
 $= \pi[1-2(x+\delta x) + (x+\delta x)^2 - (1-2x+x^2)]$
 $= \pi[1-2x-2\delta x + x^2+2x\delta x+\delta x^2 - 1+2x-x^2]$
 $= \pi(2x\delta x - 2\delta x + \delta x^2)$
 $= 2\pi(x-1)\delta x$ (ignoring δx^2 as too small).

$\therefore$ a small volume of shell is given by
 $\delta V = \delta A \cdot h$ where $h = (3-x^2) - (x^2-x)$
 $= 3+x-2x^2$

$\therefore \delta V = 2\pi(x-1)(3+x-2x^2)\delta x$
 $= 2\pi(3x+x^2-2x^3-3-x+2x^2)\delta x$
 $= 2\pi(-3+2x+3x^2-2x^3)\delta x$

$\therefore$ Volume of the solid is given by

$V = \sum \delta V$
 $= \lim_{\delta x \rightarrow 0} \sum_{x=1}^{\frac{3}{2}} 2\pi(-3+2x+3x^2-2x^3)\delta x$
 $= 2\pi \int_1^{\frac{3}{2}} (-3+2x+3x^2-2x^3) dx$

MARKING

COMMENTS

p7

① correct value for x

① area of annulus δA

alternatively:
 1 mark: volume of typical shell
 1 mark: correct limits
 1 mark: integration

① correct h leading to δV

① correct summing leading to integral

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iii) $\therefore V = 2\pi \left| \left[-3x + x^2 + x^3 - \frac{1}{2}x^4 \right]_{-1}^1 \right|$
 $= 2\pi \left| \left[(-3+1+1-\frac{1}{2}) - (-3+1-1-\frac{1}{2}) \right] \right|$
 $= 2\pi \left| -1\frac{1}{2} - 2\frac{1}{2} \right|$
 $= 8\pi$

b) $\therefore \frac{x}{(x-1)^2(x-2)} = \frac{A(x-1)(x-2) + B(x-2) + C(x-1)^2}{(x-1)^2(x-2)}$

$\therefore x = A(x-1)(x-2) + B(x-2) + C(x-1)^2 \quad \text{--- ①}$

in ①: $x=1$

$x=2$

gues: $1 = B(1-2)$

gues: $2 = C(2-1)^2$

$\therefore B = -1$

$\therefore C = 2$

also, from ①: $x = A(x^2-3x+2) + B(x-2) + C(x^2-2x+1)$

equating co-efficients of x^2 :

$0 = A + C$

$\therefore A = -2$

$\therefore \frac{x}{(x-1)^2(x-2)} = \frac{-2}{(x-1)} - \frac{1}{(x+1)^2} + \frac{2}{x-2}$

$\therefore \int_0^{\frac{1}{2}} \frac{x}{(x-1)^2(x-2)} dx$

$= \int_0^{\frac{1}{2}} \left(\frac{2}{x-2} - \frac{2}{x-1} - \frac{1}{(x-1)^2} \right) dx$

$= \int_0^{\frac{1}{2}} \left(\frac{-2}{2-x} + \frac{2}{1-x} - \frac{1}{(x-1)^2} \right) dx$

$= \left[-2 \ln(2-x) \cdot (-1) + 2 \ln(1-x) \cdot (-1) + \frac{1}{x-1} \right]_0^{\frac{1}{2}}$

$= \left[2 \ln(2-x) - 2 \ln(1-x) + \frac{1}{x-1} \right]_0^{\frac{1}{2}}$

$= 2 \ln \frac{3}{2} + 2 \ln \frac{1}{2} - 2 - (2 \ln 2 - 2 \ln 1 - 1)$

$= 2 \ln \frac{3}{2} + 2 \ln 2 - 2 - 2 \ln 2 + 0 + 1$

$= 2 \ln \left(\frac{3}{2} \right) - 1$

c) $\cos 2x = c = \sin 3x$ c constant

$\therefore \sin 3x = c$

or $\cos \left(\frac{\pi}{2} - 3x \right) = c$

$\frac{\pi}{2} - 3x = \cos^{-1}(c) + 2\pi n$ $n=0, \pm 1, \pm 2, \dots$

MARKING

COMMENTS

P

① correct integration
 ① correct answer

• Full marks only for correct solution.

① subst to find B, C (or any other method)

① equating co-efs to find A.

① correct rearrangement to get to integration

① correct subst to show answer.

$$c) \text{ (cont)} \quad \therefore -3x = -\frac{\pi}{2} + 2\pi n + \cos^{-1}(c)$$

$$\text{or } 3x = \frac{\pi}{2} - 2\pi n - \cos^{-1}(c)$$

$$\text{but } \cos 2x = c \quad \text{also,}$$

$$\text{so } 2x = \cos^{-1}(c)$$

$$\therefore 3x = \frac{\pi}{2} - 2\pi n - 2x$$

$$5x = \left(\frac{1-4n}{2}\right)\pi$$

$$\therefore x = \left(\frac{1-4n}{10}\right)\pi$$

$$\text{for } 0 \leq x \leq \frac{\pi}{2}, \text{ we get } \quad (\text{using } n=0, n=-1)$$

$$x = \frac{\pi}{10}, \frac{\pi}{2}$$

$$d) \text{ let } \tan^{-1} 3x = \theta \text{ and } \tan^{-1} 2x = \phi$$

$$\therefore \tan \theta = 3x \quad \tan \phi = 2x$$

$$\text{for } \tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}(3x) - \tan^{-1}(2x)$$

$$= \theta - \phi$$

taking tan of both sides:

$$\tan(\tan^{-1}\frac{1}{5}) = \tan(\theta - \phi)$$

$$\therefore \frac{1}{5} = \frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi}$$

$$= \frac{3x - 2x}{1 + 3x \cdot 2x}$$

$$1 + 6x^2 = 5x$$

$$\text{or } 0 = 6x^2 - 5x + 1$$

$$= 6x^2 - 3x - 2x + 1$$

$$= 3x(2x-1) - 1(2x-1)$$

$$= (2x-1)(3x-1)$$

$$\therefore x = \frac{1}{2}, \frac{1}{3}$$

QUESTION 5:

$$a) \quad \alpha + \beta + \gamma = -4$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 2$$

$$\alpha\beta\gamma = 3$$

$$i) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= (-4)^2 - 2(2)$$

$$= 12$$

MARKING

COMMENTS

① correct setup of problem (any method)

① correct solutions in range.

① correct use of tan

① forms quadratic

① correct answers

• some had an incorrect squares expansion!

① answer

$$ii) \text{ for roots } x = 1 - \alpha \Rightarrow \alpha = (1 - x)$$

$$\therefore (1-x) \text{ in eqn gives:}$$

$$(1-x)^3 + 4(1-x)^2 + 2(1-x) - 3 = 0$$

$$1 - 3x + 3x^2 - x^3 + 4 - 8x + 4x^2 + 2 - 2x - 3 = 0$$

$$\therefore 4 - 13x + 7x^2 - x^3 = 0$$

$$\text{or } x^3 - 7x^2 + 13x - 4 = 0$$

$$iii) \text{ for roots } x = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{x}$$

$$\therefore \left(\frac{1}{x}\right)^3 + 4\left(\frac{1}{x}\right)^2 + 2\left(\frac{1}{x}\right) - 3 = 0$$

$$x^3: \quad 1 + 4x + 2x^2 - 3x^3 = 0$$

$$\text{or } 3x^3 - 2x^2 - 4x - 1 = 0$$

$$b) \text{ let } \alpha \text{ be the root of multiplicity 3, then } P(\alpha) = P'(\alpha) = P''(\alpha) = 0.$$

$$\therefore P'(\alpha) = 32\alpha^3 - 75\alpha^2 + 54\alpha - 11$$

$$P''(\alpha) = 96\alpha^2 - 150\alpha + 54$$

$$\text{if } P(\alpha) = 0, \alpha \text{ is the soln to } 0 = 96\alpha^2 - 150\alpha + 54$$

$$\text{or } 0 = 48\alpha^2 - 75\alpha + 27$$

$$\therefore \alpha = \frac{75 \pm \sqrt{2625 - 4 \cdot 48 \cdot 27}}{96}$$

$$= \frac{75 \pm \sqrt{441}}{96}$$

$$= \frac{75 \pm 21}{96}$$

$$= 1, \frac{9}{16}$$

$$\text{now, } P'(1) = 32 - 75 + 54 - 11$$

$$= 0$$

$$\text{and } P(1) = 8 - 25 + 27 - 11 + 1$$

$$= 0$$

$$\therefore (x-1)^3 \text{ is a factor of } P(x)$$

$$\text{so } \alpha = 1 \text{ is the tripple root.}$$

$$\text{Also, } \alpha^3 \beta = \frac{1}{8}$$

$$\therefore \beta = \frac{1}{8} \text{ is the other root.}$$

$$c) \text{ let } \alpha, \beta, \gamma \text{ and } \delta \text{ be the roots of the equation}$$

$$i) \therefore x = \alpha + 1 \text{ is a root of the reqd eqn}$$

$$\text{so } \alpha = x - 1$$

$$\therefore (x-1)^4 + 4(x-1)^3 + 5(x-1)^2 + 2(x-1) - 20 = 0$$

MARKING

COMMENTS

① correct setup with roots
① correct eqn.

• many simple algebraic errors

① setup with roots
① correct eqn.

① set up problem with $P''(\alpha) = 0$.

• several used $P'''(\alpha) = 0$
• many did not understand the implications for a root with multiplicity!

① correct possibilities for tripple root.

① correct tripple root with reasons

① other root

• need to explicitly state the other root. $(8x-1)$ as a factor implies a root of $x = \frac{1}{8}$.

① correct subst for root.

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MARKING

COMMENTS

p11

i) (cont):

$$(x^4 - 4x^3 + 6x^2 - 4x + 1) + (4x^3 - 12x^2 + 12x - 4) + (5x^2 - 10x + 5) + 2x - 2 = 0$$

$$\therefore x^4 - x^2 - 20 = 0 \quad \text{as reqd.}$$

ii) now $x^4 - x^2 - 20 = 0$

$$(x^2 - 5)(x^2 + 4) = 0$$

$$\therefore x^2 = 5, -4$$

$$\therefore x = \pm\sqrt{5}, \pm 2i$$

 $\therefore$ the roots of $x^4 + 4x^3 + 5x^2 + 2x - 20 = 0$
are given by $\alpha = x - 1$
 $\therefore$ roots are $-1 + \sqrt{5}, -1 - \sqrt{5}, -1 + 2i, -1 - 2i$

QUESTION 6:

a) i) $x_p = 0.5$ $y_p = 0.7$

$$= 5 \cos \theta$$

$$= 3 \sin \theta$$

$$\text{ii) } \therefore \frac{x^2}{5} = \cos^2 \theta \quad \text{and} \quad \frac{y^2}{9} = \sin^2 \theta$$

$$\frac{x^2}{25} = \cos^2 \theta \quad \frac{y^2}{9} = \sin^2 \theta$$

$$\therefore \frac{x^2}{25} + \frac{y^2}{9} = \cos^2 \theta + \sin^2 \theta$$

$$\text{so } \frac{x^2}{25} + \frac{y^2}{9} = 1 \quad \text{is the cartesian eqn.}$$

iii) normal to an ellipse is given by $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ where $a=5, b=3$

$$\theta = \frac{\pi}{3}$$

$$\therefore \frac{5x}{\cos \frac{\pi}{3}} - \frac{3y}{\sin \frac{\pi}{3}} = 25 - 9$$

$$10x - \frac{6y}{\sqrt{3}} = 16$$

$$\therefore 10\sqrt{3}x - 6y - 16\sqrt{3} = 0 \quad \text{is eqn.}$$

b) i) $S'(x) = \frac{d}{dx} \left(\frac{1}{2}(e^x - e^{-x}) \right)$

$$= \frac{1}{2}(e^x + e^{-x})$$

$$= C(x)$$

ii) $e^x > 0$ for all x

$$e^{-x} > 0 \quad \text{for all } x$$

$$\therefore e^x + e^{-x} > 0 \quad \text{for all } x$$

$$\therefore C(x) > 0 \quad \text{for all } x$$

$$\text{ie } S'(x) > 0 \quad \text{for all } x$$

 $\Rightarrow S(x)$ is monotonically increasing

① correct alg.
to soln.

• many errors expanding
this

① correct roots
for $x^4 - x^2 - 20 = 0$

① correct
roots for orig.
eqn.

• many had trouble linking
the roots back to the
original with $\alpha = x - 1$

①
① each
answer

• link back to the definition
given in the diagram. This
is the starting point, and many
missed it.

① correct
eqn.

• "eliminate θ " $\Rightarrow$ show how
this happens, don't just write
the equation down!

① correct subst
in formula

① correct eqn
(any form)

• many found tangent instead
of normal.

• put your answer into one of
the standard simplest forms - many
left their answer unfinished

① set out
clearly.

• asked to show $\Rightarrow$ give
reasons why $S'(x) > 0$. Just
stating it earns no marks.

① correct
reasoning.

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MARKING

COMMENTS

p1

$$\begin{aligned} \text{iii) } [C(x)]^2 &= \left[\frac{1}{2}(e^x + e^{-x}) \right]^2 \\ &= \frac{1}{4}(e^{2x} + 2e^x e^{-x} + e^{-2x}) \\ &= \frac{1}{4}(e^{2x} + e^{-2x} + 2) \end{aligned}$$

$$\begin{aligned} 1 + [S(x)]^2 &= 1 + \left[\frac{1}{2}(e^x - e^{-x}) \right]^2 \\ &= 1 + \frac{1}{4}(e^{2x} - 2e^x e^{-x} + e^{-2x}) \\ &= \frac{1}{4}(4 + e^{2x} - 2 + e^{-2x}) \\ &= \frac{1}{4}(e^{2x} + e^{-2x} + 2) \\ &= [C(x)]^2 \quad \text{from above} \end{aligned}$$

iv) as $S(x)$ is monotonically increasing,
each x must produce a unique y value

 $\Rightarrow S(x)$ has a 1-1 correspondence

 $\therefore S^{-1}(x)$ exists for all values of x .
v) $y = S^{-1}(x)$

$$\therefore S(y) = x$$

$$\therefore \frac{dy}{dx} = S'(y)$$

$$= C(y)$$

$$= \sqrt{1 + [S(y)]^2}$$

$$= \sqrt{1 + x^2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1 + x^2}}$$

$$\text{vi) } \therefore y = \int \frac{dx}{\sqrt{1 + x^2}}$$

$$\text{let } x = \tan \theta \quad \therefore dx = \sec^2 \theta d\theta$$

$$= \int \frac{\sec^2 \theta d\theta}{\sqrt{1 + \tan^2 \theta}}$$

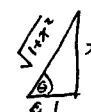
$$= \int \frac{\sec^2 \theta d\theta}{\sec \theta}$$

$$= \int \sec \theta d\theta$$

$$= \int \frac{\sec \theta (\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} d\theta$$

$$= \ln(\sec \theta + \tan \theta) + C$$

$$y = \ln(x + \sqrt{1 + x^2}) + C$$



① inverse rules
to give $\frac{dy}{dx}$
correctly
① correct subst
to formula.

① reduction
to $\int \sec \theta d\theta$

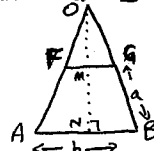
① correct $\int$
of $\sec \theta$

① correct subst
to give y in terms
of x .

• very few picked up the link
to $S(x)$ and $C(x)$ in the previ
parts, so many futile at
a simple problem. Look
linked parts like this one -
make the solution simple

• The question is to show a
relationship $\Rightarrow$ not use
the standard integral table. If
integration is the question
it's not part of something
bigger.

QUESTION 7:

a) i) In base ΔOAB :

$$GB = a$$

$$OB = r$$

$$NB = \frac{b}{2}$$

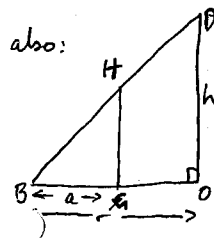
$$OG = r - a$$

① correct
setup of
variables

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a) (cont) $\therefore \frac{MB}{OB} = \frac{NB}{OB}$
 or $MB = \frac{NB \cdot OB}{OB}$
 $= \frac{b \cdot (r-a)}{r}$

$\therefore FG = 2MB$
 $= \frac{2b(r-a)}{r}$



$OB = h, OB = r$
 $GB = a$

$\therefore \frac{FG}{GB} = \frac{OB}{OB}$
 $\therefore FG = \frac{OB \cdot GB}{OB}$
 $= \frac{a \cdot h}{r}$

$\therefore V_s = FG \cdot \Delta H \cdot \Delta a$
 $= \left(\frac{r-a}{r}\right) b \cdot \left(\frac{ah}{r}\right) \Delta a$

ii) $\therefore V = \int_0^r \left(\frac{r-a}{r}\right) b \cdot \left(\frac{ah}{r}\right) da$

$= \frac{bh}{r^2} \int_0^r a(r-a) da$
 $= \frac{bh}{r^2} \int_0^r ar - a^2 da$
 $= \frac{bh}{r^2} \left[\frac{1}{2} ar^2 - \frac{1}{3} a^3 \right]_0^r$
 $= \frac{bh}{r^2} \left[\left(\frac{1}{2} r^3 - \frac{1}{3} r^3 \right) - 0 \right]$
 $= \frac{bh}{r^2} \cdot \left(\frac{1}{2} - \frac{1}{3} \right) r^3$
 $= \frac{1}{6} bhr$ as reqd.

iii) given $\angle AOB = \frac{2\pi}{n}$



ie $\theta = \frac{2\pi}{n}$
 $\therefore \frac{\theta}{2} = \frac{\pi}{n}$

$\therefore \sin \frac{\theta}{2} = \frac{b}{2} \cdot \frac{1}{r}$
 or $b = 2r \sin \frac{\theta}{2}$
 $= 2r \sin \frac{\pi}{n}$

$\therefore V = \frac{1}{6} bhr$ from (ii) above

so $V = \frac{1}{6} hr \cdot 2r \sin \frac{\pi}{n}$
 $= \frac{1}{3} hr^2 \sin \frac{\pi}{n}$

ie $V_n = \frac{1}{3} hr^2 n \sin \frac{\pi}{n}$

MARKING

COMMENTS

① correct value for FG

① correct expression for ΔH .

① reduction to correct $\int$

① correct subst to expression

① correct derivation for b.

① correct expression for V.

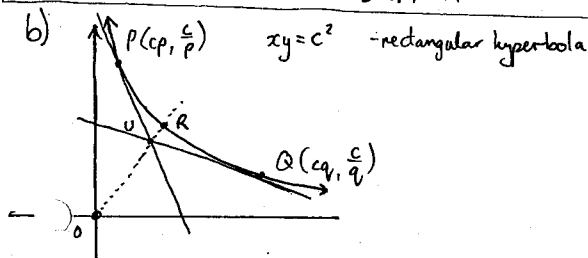
p13

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iv) (cont) $\therefore \lim_{n \rightarrow \infty} V_n = \lim_{n \rightarrow \infty} \frac{1}{3} nhr^2 \sin \frac{\pi}{n}$
 $= \lim_{n \rightarrow \infty} \frac{1}{3} hr^2 \pi \cdot \frac{\sin \frac{\pi}{n}}{\frac{\pi}{n}}$

let $x = \frac{\pi}{n}$; as $n \rightarrow \infty, \frac{\pi}{n} \rightarrow 0$

$\therefore \lim_{n \rightarrow \infty} V_n = \lim_{n \rightarrow \infty} \frac{1}{3} hr^2 \pi \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$
 $= \frac{1}{3} \pi r^2 h$



tangent at P: $x + p^2 y = 2cp$ — ①

" Q: $x + q^2 y = 2cq$ — ②

$(p^2 - q^2)y = 2c(p - q)$: ① - ②

$\therefore y_u = \frac{2c}{p+q}$

$\therefore x_u = 2cp - \frac{p^2 2c}{p+q}$

Now $m_{uv} = \frac{2c}{p+q} \cdot \frac{p+q}{2cp}$
 $= \frac{1}{pq}$

$\therefore$ eqn of OUR is $y = \frac{x}{pq}$ — ③

subst ③ in ④: $xy = c^2$ — ④
 $\frac{x^2}{pq} = c^2$

$\therefore x^2 = pq c^2$

$\therefore x_R = c\sqrt{pq}$

but R is $(cr, \frac{c}{r})$

so $Cr = c\sqrt{pq}$

$r = \sqrt{pq}$

or $r^2 = pq$

MARKING

COMMENTS

P

① limit expression
 ① correct use of limits to evaluate expression

• many incorrect uses of the limit:

① diagram showing relationships

• many students didn't draw correct diagrams!

① finding y coord of U

• not many students completed the correct relationship.

① finding x coord of U

• Some became lost after getting correct points of intersection, others made it much more complicated than it was.

① gradient of uv

① finding x_R (or y_R)

① correct relationships (either form)

QUESTION 8:

a) i) $\int_0^{\frac{2}{3}} \sqrt{4-9x^2} dx$

$$x = \frac{2}{3} \sin \theta$$

$$\therefore dx = \frac{2}{3} \cos \theta d\theta$$

when $x = \frac{2}{3}$, $\theta = \frac{\pi}{2}$
 $x = 0$, $\theta = 0$

$$= \int_0^{\frac{\pi}{2}} \sqrt{4-4\sin^2 \theta} \cdot \frac{2}{3} \cos \theta d\theta$$

$$= \frac{4}{3} \int_0^{\frac{\pi}{2}} \cos^2 \theta \cdot \cos \theta d\theta$$

$$= \frac{4}{3} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= \frac{4}{3} \int_0^{\frac{\pi}{2}} \frac{1}{2} (\cos 2\theta + 1) d\theta$$

$$= \frac{2}{3} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{2}{3} \left[\left(\frac{1}{2} \cdot 0 + \frac{\pi}{2} \right) - 0 \right]$$

$$= \frac{\pi}{3}$$

① correct
subst to θ ,
including limits

① correct reduction
to $\int \cos^2$

① correct subst
to soln.

ii) for $9x^2 + y^2 = 4$

$$y^2 = 4 - 9x^2$$

$$\therefore y = \sqrt{4-9x^2}$$

$\therefore$ pt i) gives the area in the first quadrant
 so, from symmetry, this is $\frac{1}{4}$ the reqd. area.
 $\therefore A = 4 \cdot \frac{\pi}{3}$
 $= \frac{4\pi}{3}$ sq units.

① answer.

b) i) $x = a \sin \theta$

$$\therefore dx = a \cos \theta d\theta$$

when $x = a$, $\theta = \frac{\pi}{2}$
 $x = 0$, $\theta = 0$

$$\therefore \int_0^a \sqrt{a^2 - x^2} dx$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta$$

$$= a^2 \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} \cdot \cos \theta d\theta$$

$$= a^2 \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= a^2 \int_0^{\frac{\pi}{2}} \frac{1}{2} (\cos 2\theta + 1) d\theta$$

$$= a^2 \left[\frac{1}{4} \sin 2\theta - \frac{\theta}{2} \right]_0^{\frac{\pi}{2}}$$

$$= a^2 \left[\left(0 + \frac{\pi}{4} \right) - 0 \right]$$

$$= \frac{a^2 \pi}{4}$$

① correct
subst inc
limits

① correct
integration to
solution

ii) from $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$b^2 x^2 + a^2 y^2 = a^2 b^2$$

$$a^2 y^2 = a^2 b^2 - b^2 x^2$$

$$\therefore y^2 = b^2 - \frac{b^2}{a^2} x^2$$

$$\therefore y = \sqrt{b^2 - \frac{b^2}{a^2} x^2}$$

$$= \sqrt{\frac{b^2 a^2 - b^2 x^2}{a^2}}$$

$$= \frac{b}{a} \sqrt{a^2 - x^2}$$

$\therefore$ area of 1st quadrant is

$$A_1 = \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= \frac{b}{a} \cdot \frac{\pi a^2}{4} \quad \text{from (i)}$$

$$= \frac{\pi ab}{4}$$

$\therefore$ total area (from symmetry)

$$A = 4 \cdot \frac{\pi ab}{4}$$

$$= \pi ab$$

① reducing
equation to
std. form

① correct
reasoning to
soln.

c) $\Delta V = A \Delta h$ where A is the area of the ellipse at height h .

$\therefore$ from (b) above:

$$A = \pi ab \lambda^2$$

(as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2$ becomes
 $\frac{x^2}{a^2 \lambda^2} + \frac{y^2}{b^2 \lambda^2} = 1$)

$$\therefore \Delta V = \pi ab \lambda^2 \Delta h$$

$$\therefore V = \int_0^H \pi ab \left(1 - \frac{h^2}{H^2} \right)^2 dh$$

$$= \pi ab \int_0^H \left(1 - \frac{2h^2}{H^2} + \frac{h^4}{H^4} \right) dh$$

$$= \pi ab \left[h - \frac{2}{3} \frac{h^3}{H^2} + \frac{h^5}{5H^4} \right]_0^H$$

$$= \pi ab \left[\left(H - \frac{2}{3} \frac{H^3}{H^2} + \frac{H^5}{5H^4} \right) - 0 \right]$$

$$= \pi ab \left[\frac{15H - 10H + 3H}{15} \right]$$

$$= \frac{8\pi ab H}{15} \quad \text{as reqd.}$$

① correct
deduction of A

① correct
expression for
 V in terms of
 h 's.

① correct $\int$
leading to
soln

• wrong λ , but correct
method, gained 1 mark

d) $x^2 - (2\cos\theta)x + 1 = 0$

i) $x^2 - (2\cos\theta)x + \cos^2\theta = -1 + \cos^2\theta$

$\therefore (x - \cos\theta)^2 = -\sin^2\theta$

$\therefore x - \cos\theta = \pm i \sin\theta$

$\therefore x = \cos\theta \pm i \sin\theta$

$\therefore \alpha = \cos\theta + i \sin\theta \quad \beta = \cos\theta - i \sin\theta$

ii) $\alpha = \cos\theta$

$\therefore \alpha^{10} = (\cos\theta)^{10}$

$= \cos 10\theta$ by de Moivre's Theorem

similarly $\beta = \overline{\cos\theta}$

so $\beta^{10} = (\overline{\cos\theta})^{10}$

$= \overline{\cos 10\theta}$

$\therefore \alpha^{10} + \beta^{10} = \cos 10\theta + \overline{\cos 10\theta}$

$= \cos 10\theta + i \sin 10\theta + \cos 10\theta - i \sin 10\theta$

$= 2 \cos 10\theta$ as reqd.

both
① answers

① correct use
of de Moivre's
Theorem

① correct
use of $\overline{\cos\theta}$

① correct
algebra to
soln.